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# UNIT 2

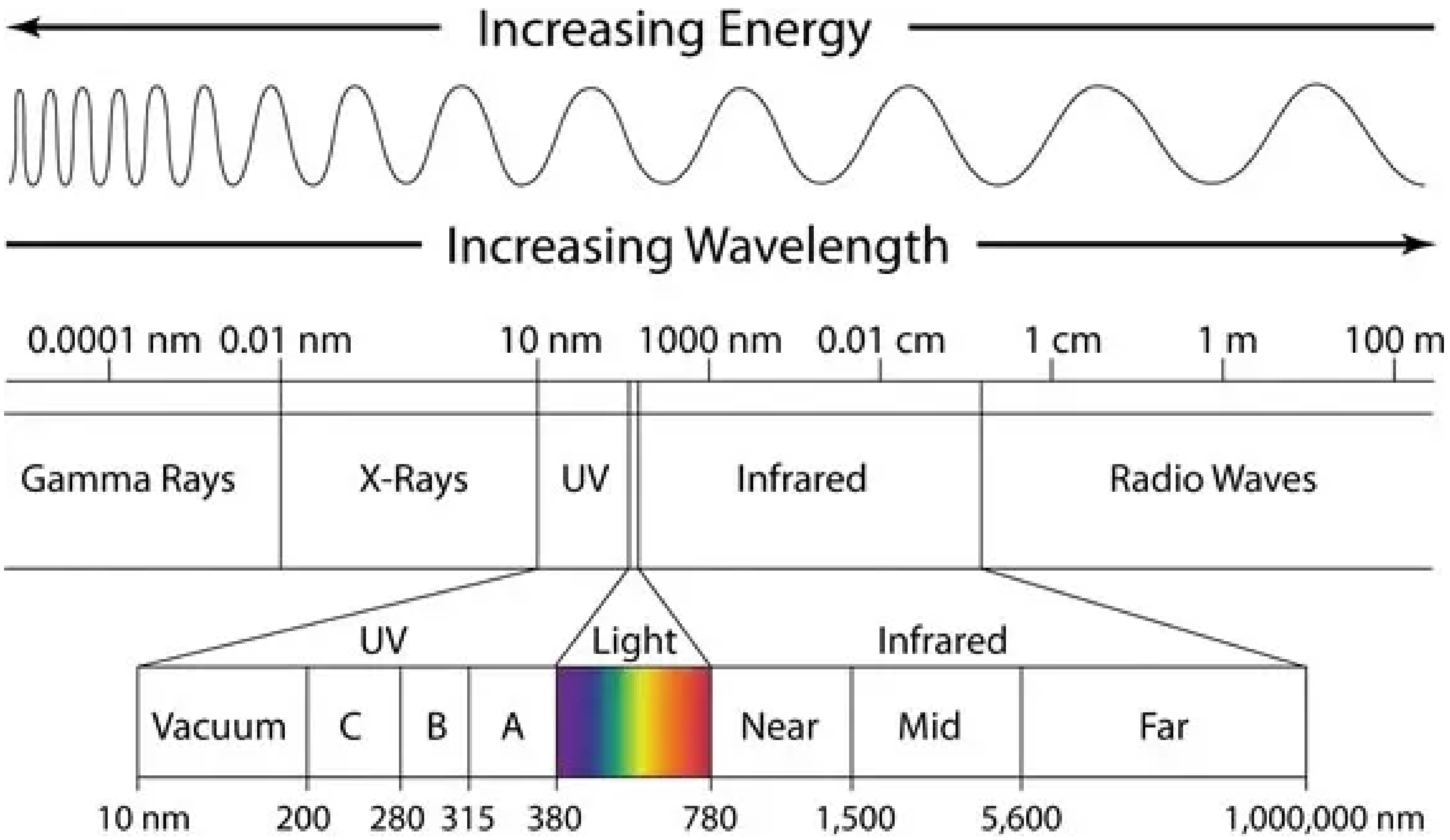
# WAVE PROPERTIES OF LIGHT

By
**Dr. Madhuri D. R**
Assistant Professor
School of Humanities and Engineering Sciences
MITAOE, Alandi, Pune

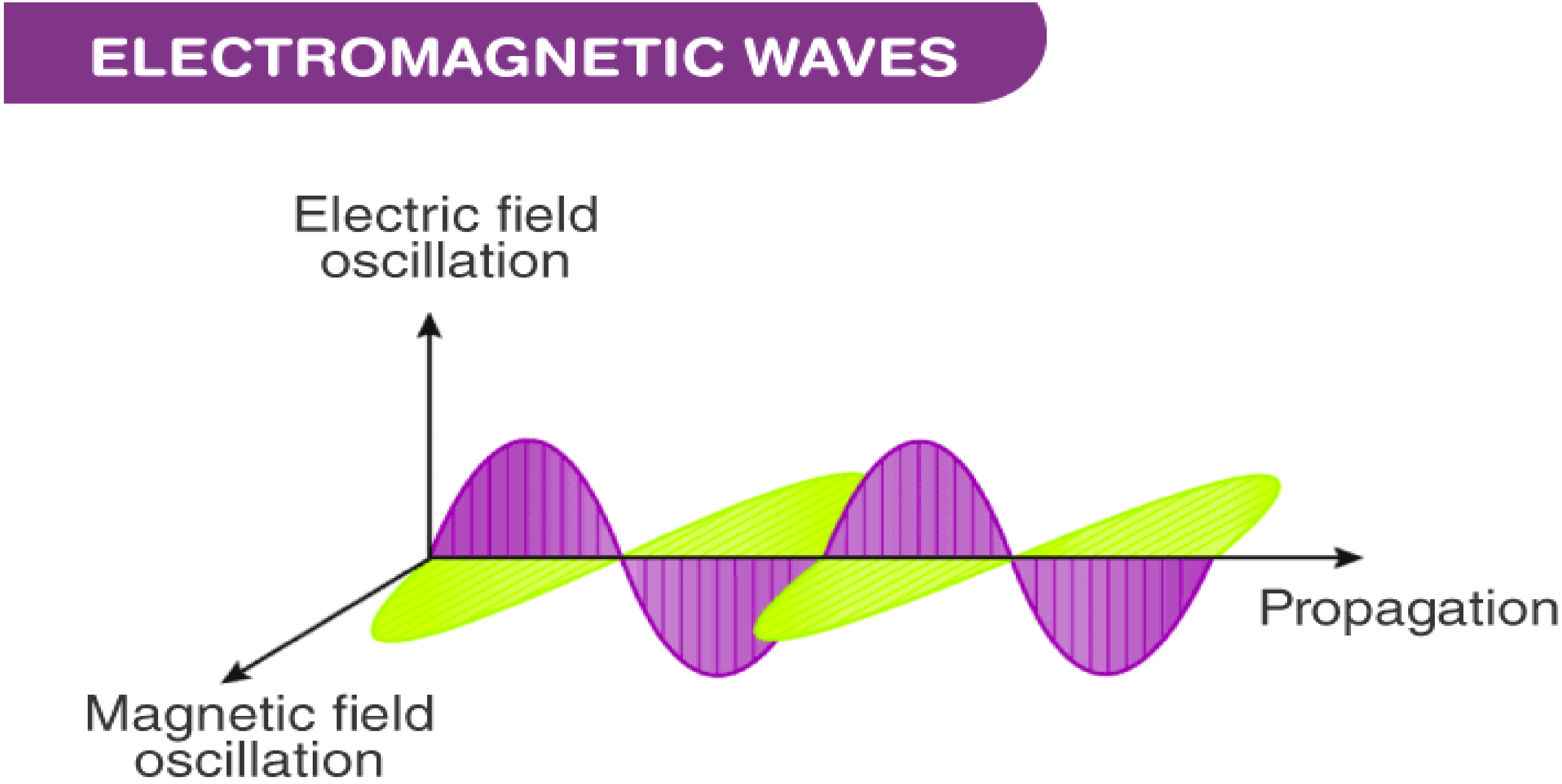
# Contents

- Particle nature and wave-nature with examples of wave and particle behavior light
- Introduction to wave nature, Concept of thin film
- Stokes law of phase-change on reflection from a thin film
- Thin film interference
- Interference in films of uniform and non-uniform thickness (with derivation)
- Newton Ring Experiment
- Basics of polarization of electromagnetic wave.

# Electromagnetic spectra



# Electric and magnetic vectors are perpendicular to each other



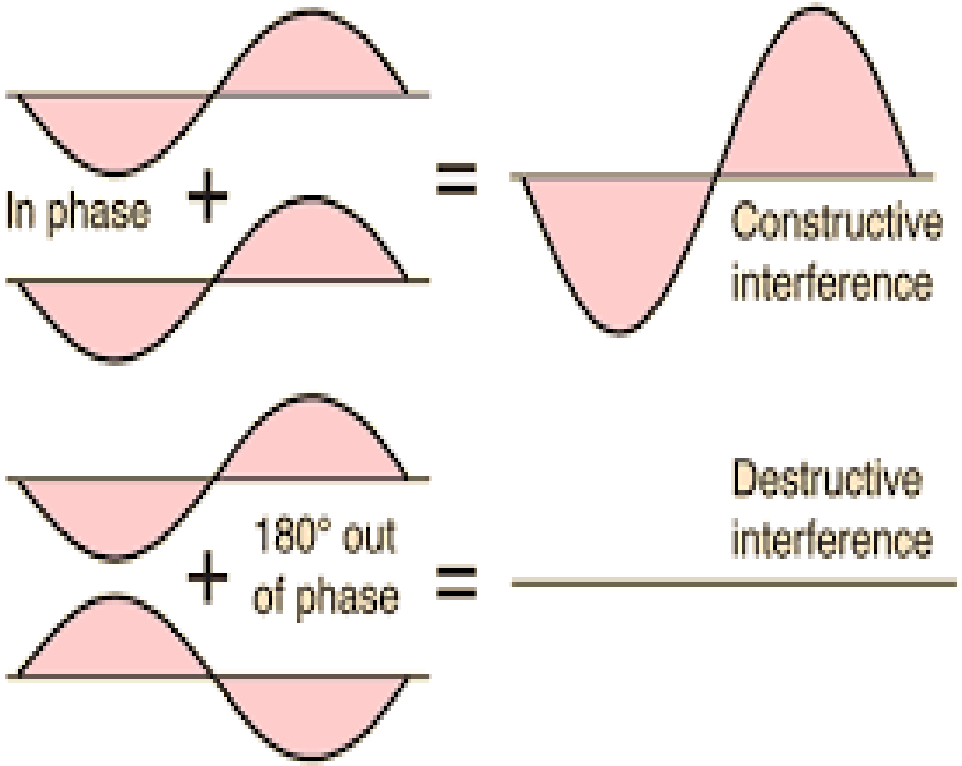
# Law of superposition

When two or more waves overlap in space, the resulting disturbance is equal to the vector sum of the individual disturbance.

Amplitude of resultant wave = A+A= 2A

Intensity of resultant wave I $\alpha$ $A^2$

I $\alpha$ $(2A)^2 = 4A^2 = 4I$



# Interference

**Interference is a phenomenon in which two or more waves superpose to form a resultant wave of greater, lower or of the same amplitude.**

**OR**

**Redistribution of light intensity due to superposition of two or more waves.**

## Conditions to get interference pattern

- The two sources must be **coherent** (phase difference between waves are constant or same).
- The two sources must be **monochromatic** (same wavelength).
- Interfering waves should be of **equal amplitude.**

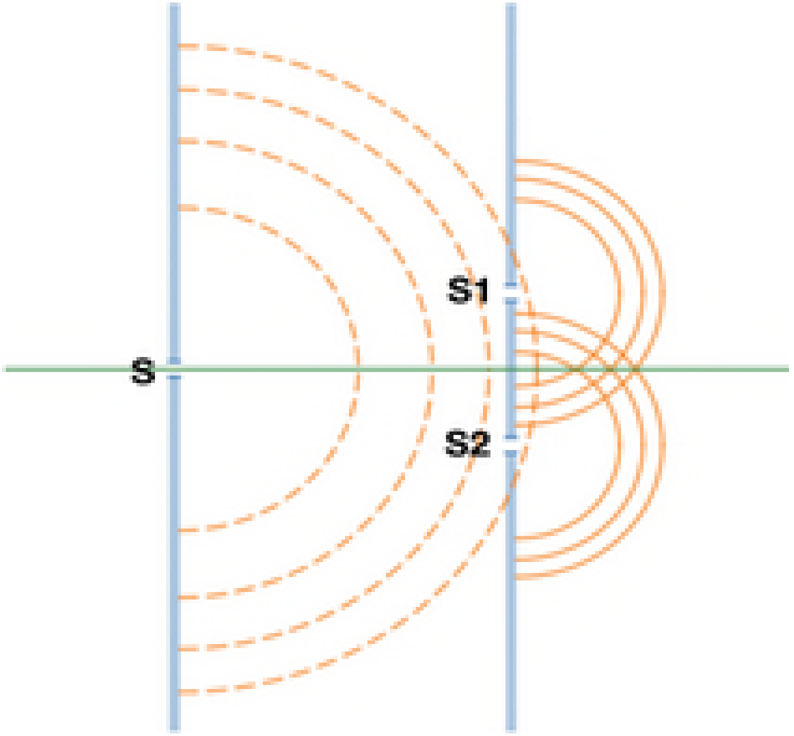
# Techniques of obtain coherent waves

## 1. Division of Wavefront

In this class of interference, a single wavefront produces two waves of equal phase of which superposition creates the interference pattern.

A wavefront is an imaginary surface representing corresponding points of a wave that vibrate simultaneously.

A wavefront is defined as a surface over which the phase of the wave is constant.

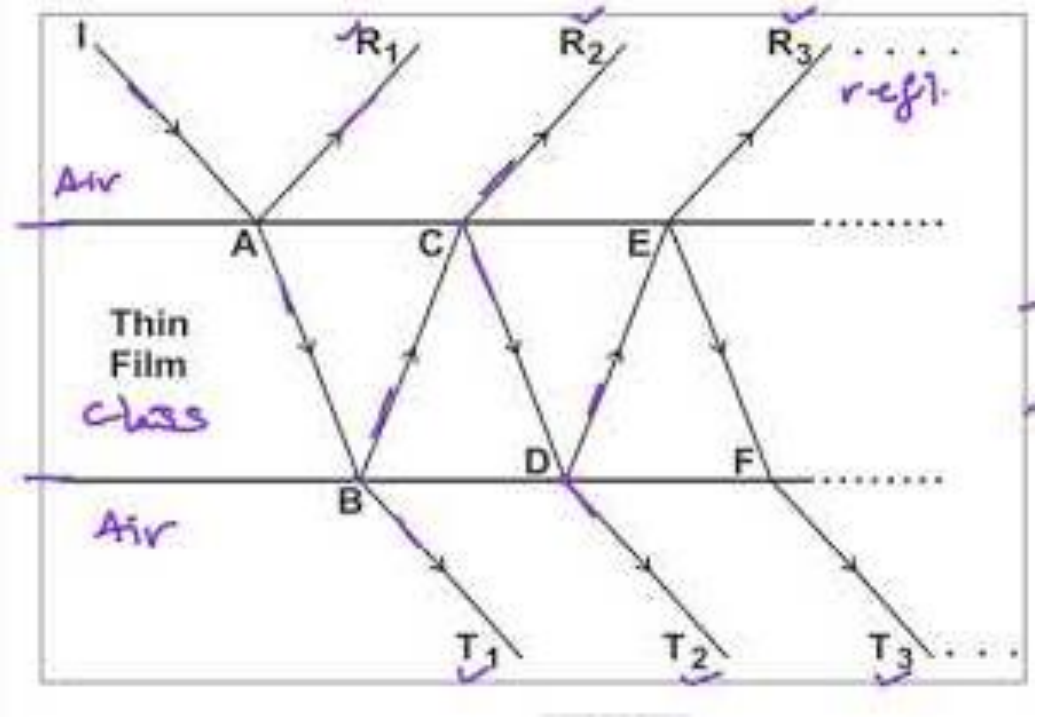



Young's double slit experiment,
Biprism.

## 2. Division of Amplitude

If a wave of amplitude A falls on a dividing surface, then a part of it is reflected and a part is transmitted.

A method of creating coherent light by splitting the amplitude of an incoming light beam into multiple parts through partial reflection or refraction, allowing these separated parts to travel different paths.



## Constructive interference

- When two waves meet at a point in phase, the resultant amplitude is addition of two individual amplitudes. This leads to increase in intensity as intensity is directly proportional to square of amplitude.

Path difference:
$\Delta = n\lambda$

## Destructive interference

- When two waves meet at a point out of phase, the resultant amplitude is difference between two individual amplitudes. This leads to decrease in intensity and is called destructive interference.

Path difference:
$\Delta = (2n+1)\lambda/2$

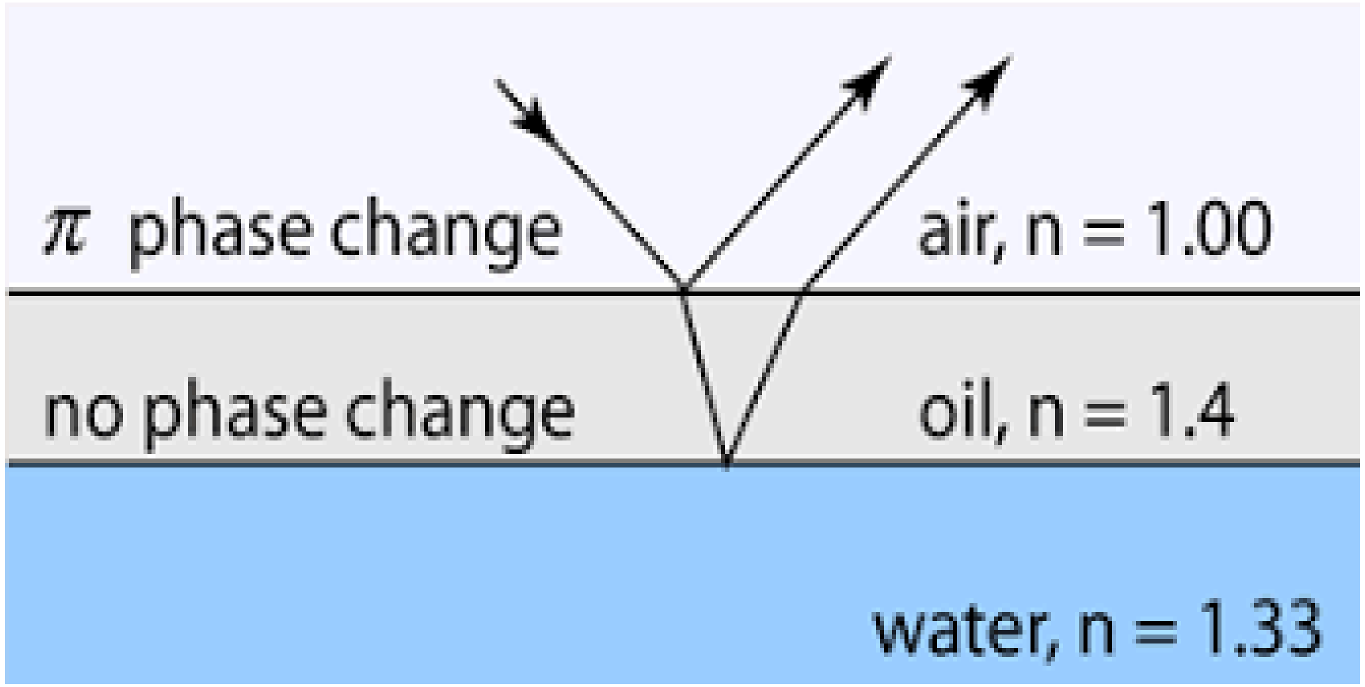
Where n is an integer. n = 0, 1, 2,…

# Stoke's law

When a light wave is reflected from the surface of an optically denser medium, it suffers a phase change of $\boldsymbol{\pi}$ or path change of $\boldsymbol{\lambda/2}$, but it suffers no change in phase or path when reflected at the surface of optically rarer medium.

**Path changes by λ/2**

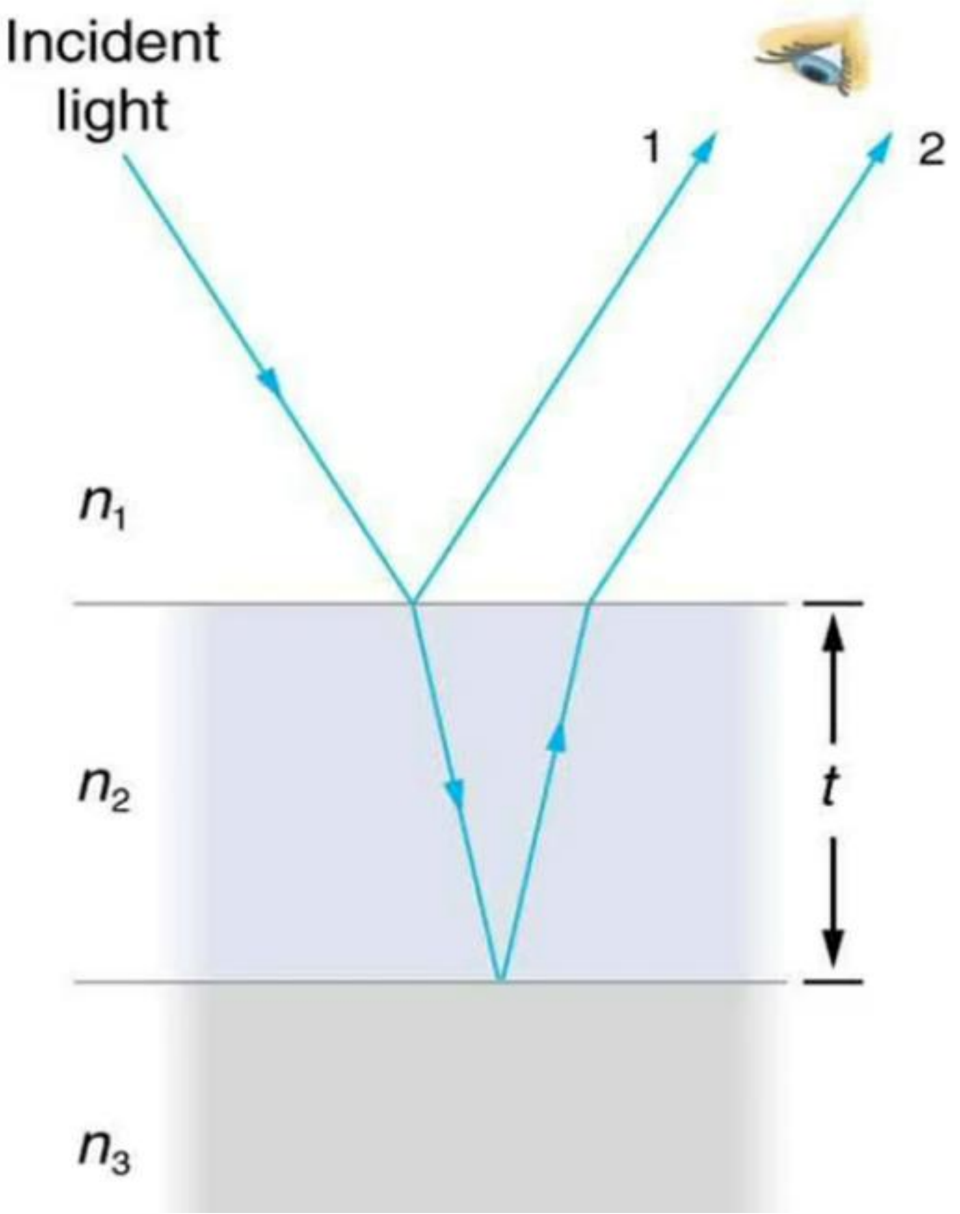
**No path changes**



# THIN FILMS

**Thin film interference** occurs when light reflects from both boundaries of a thin film of material.

**An optical medium is said to be a thin film <u>when the thickness of the film is about 1 wavelength of light in visible region.</u>**



# Interference in Thin film of uniform thickness

The beautiful color produced by a thin film of thickness of oil on the surface of water and also by the film of soap bubble can be explained by studying the interference phenomenon in thin films.

- Consider a transparent film of thickness t and refractive index $\mu > 1$
- Let AB is a monochromatic light ray incident on top surface of film at point B. Some part of light ray get reflected along $R_1$ and some part get transmitted along BC. At point C again light ray get reflected along CD and emerge out as $R_2$ and transmitted light ray goes along $T_1$.
- Reflected ray $R_1$ and $R_2$ superpose with each other with the help of lenses and interference pattern of dark and bright fringes produced.

# Derivation:

$R_1$ and $R_2$ both are coherent because they are produced from ray AB.

**The geometrical path difference produced between $R_1$ and $R_2$ is,**

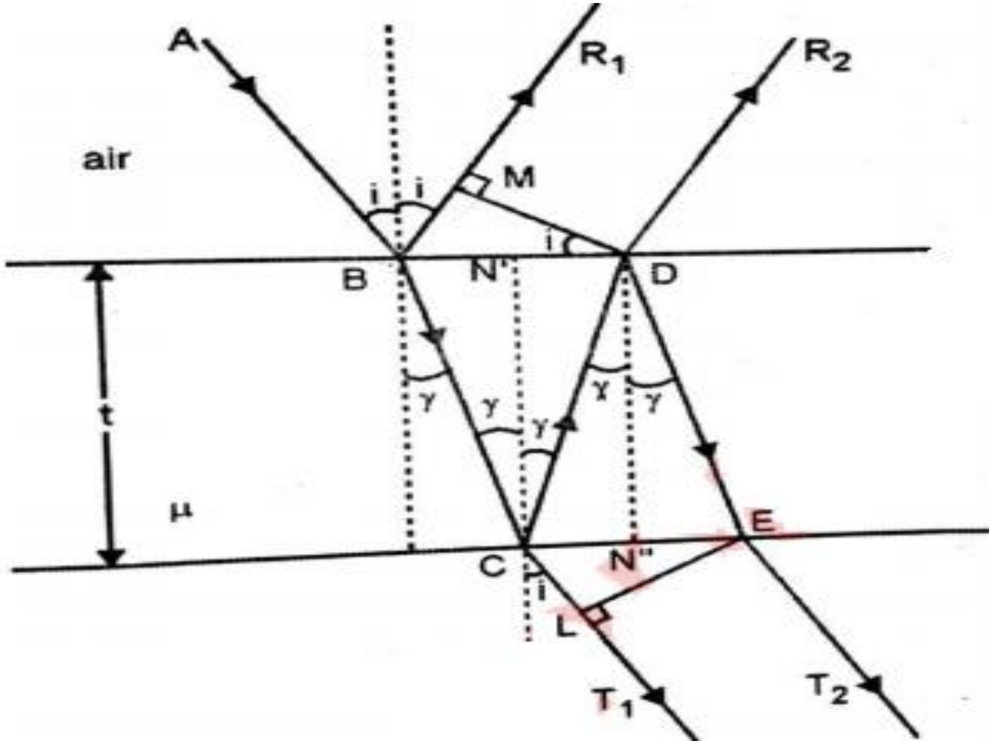
**$\Delta = \mu\,(BC + CD) - BM$ ........................(1)**

**In triangle BDM**

**$\sin i = BM/BD$.   $BM = BD \sin i = (BN+ND) \sin i$**

**As BN=ND, BD = 2BN. So above $eq^n$ can be written as**

**$BM = 2BN \sin i$ ..............................(2)**




**In triangle BCN and DCN   $BC = CD = t/\cos r$ ...................(3)**

**In ΔBCN, $\tan r = BN/CN$ . $BN = t \tan r$.   put this value of BN in equation (2)**

**$BM = (2t) \tan r \cdot \sin i$.   By using Snells law, $\mu = \sin i / \sin r$.   $\sin i = \mu \sin r$**

**$BM = 2t\,\mu \sin r\,(\sin r / \cos r)$.   $BM = 2\mu t\,(\sin^2 r/\cos r)$ ........................(4)**

**Substituting (4) and (3) in equation (1)**

**$\Delta = \mu\,(t/\cos r + t/\cos r) - 2\mu t\,(\sin^2 r/\cos r)$**

**$= (2\mu t/\cos r) \cdot (1 - \sin^2 r)$.   Where $(1 - \sin^2 r) = \cos^2 r$**

**$\boxed{\Delta = 2\mu t \cos r.}$ Is the path difference between two reflected light waves**

# Path difference in thin film due to reflection:

When light rays reflected from the surface of a denser medium path diffrence changes by $\lambda/2$ and phase changes by $\pi$

$$\boxed{\Delta = 2\mu t \cos r \pm (\lambda/2)} \quad ................... 6$$

## 1. Conditon for constructive interference/ maxima/ bright fringe:

Path difference between two waves is integral multiple $\lambda$

$$\Delta = n\lambda \quad .......................... 7$$

Equating equations 6 & 7 we can write

$$2\mu t \cos r \pm (\lambda/2) = n\lambda$$

$$\boxed{2\mu t \cos r = (2n \pm 1)\, \lambda/2} \quad ........................... 8$$

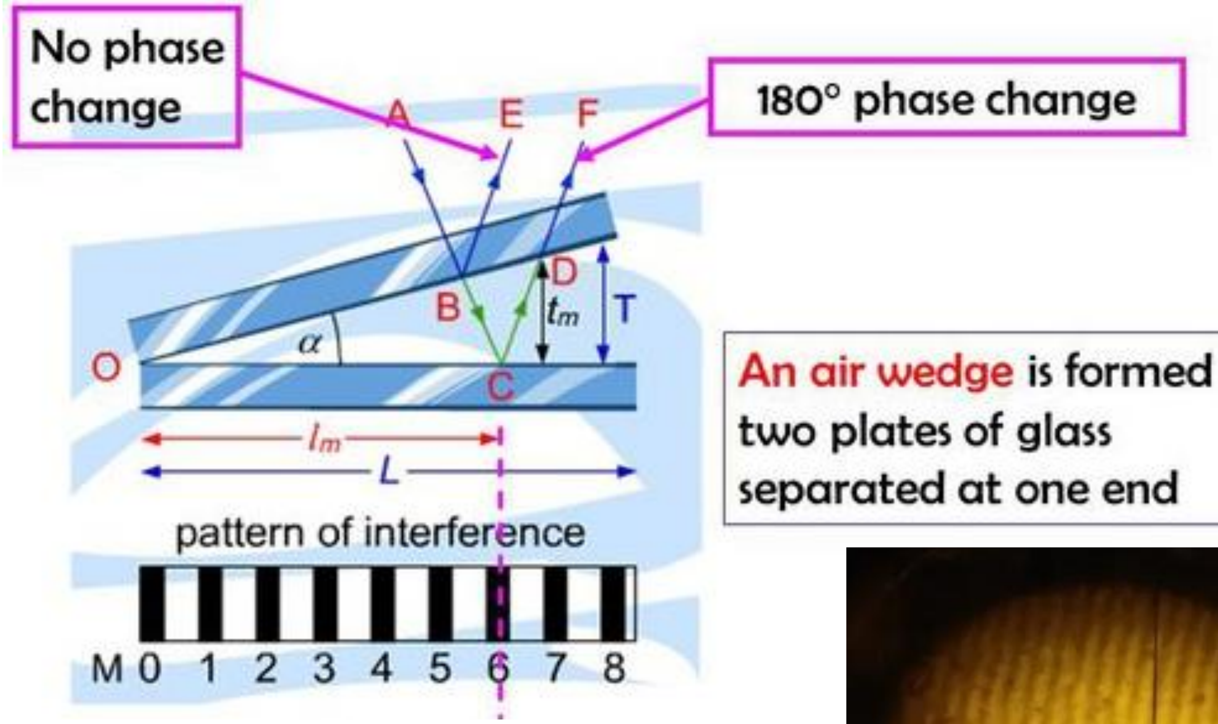
## 2. Condition for destructive interference/ minima/ dark fringe:

Path difference between two waves is odd integral multiple of $\lambda$

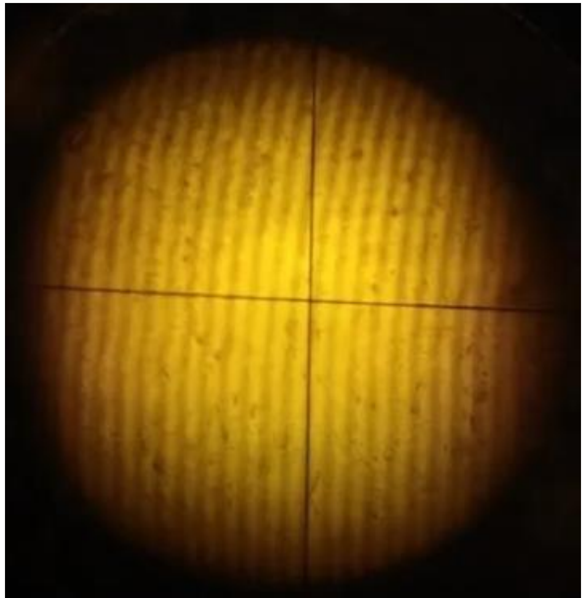
$\Delta = (2n\pm1)\, \lambda/2$ ........................ 9   Equating the equations 6 and 9, we get

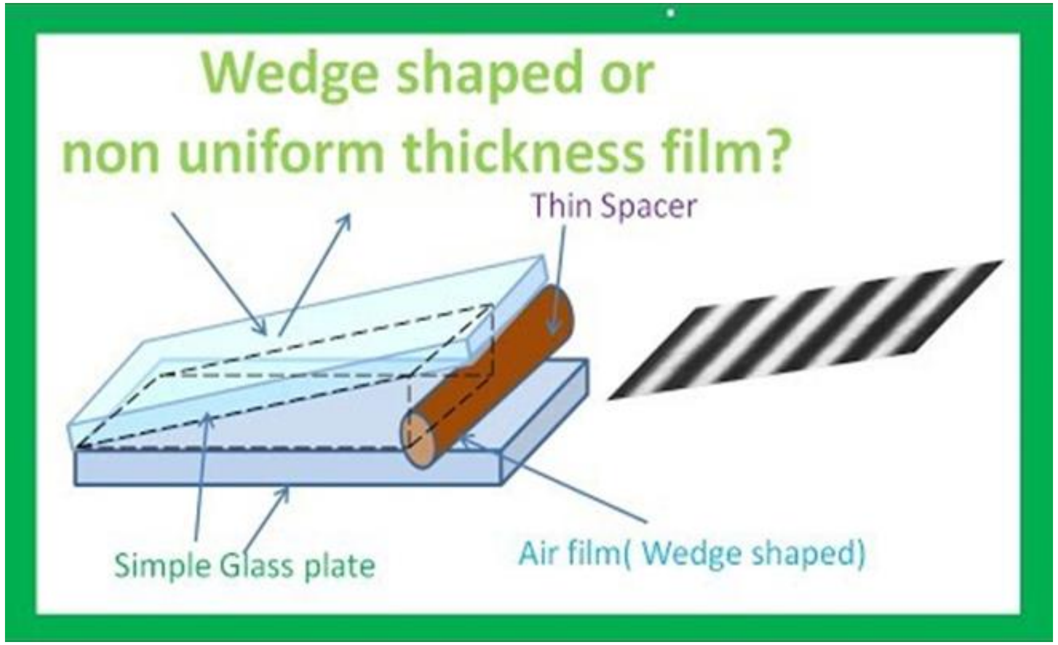
$$\boxed{2\mu t \cos r = n\lambda} \quad ...................... 10$$

# Interference in Thin film of non-uniform thickness (Wedge Shaped Film)




An air wedge is formed by two plates of glass separated at one end





- When this film is illuminated by the monochromatic light and the reflected light is viewed from the upper surface, alternate dark and bright fringes are observed.
- This is due to the interference of the reflected rays.
- The interfering rays BR and $DR_1$ are not parallel but appear to diverge from each other.

Let us calculate the path difference between these two rays.

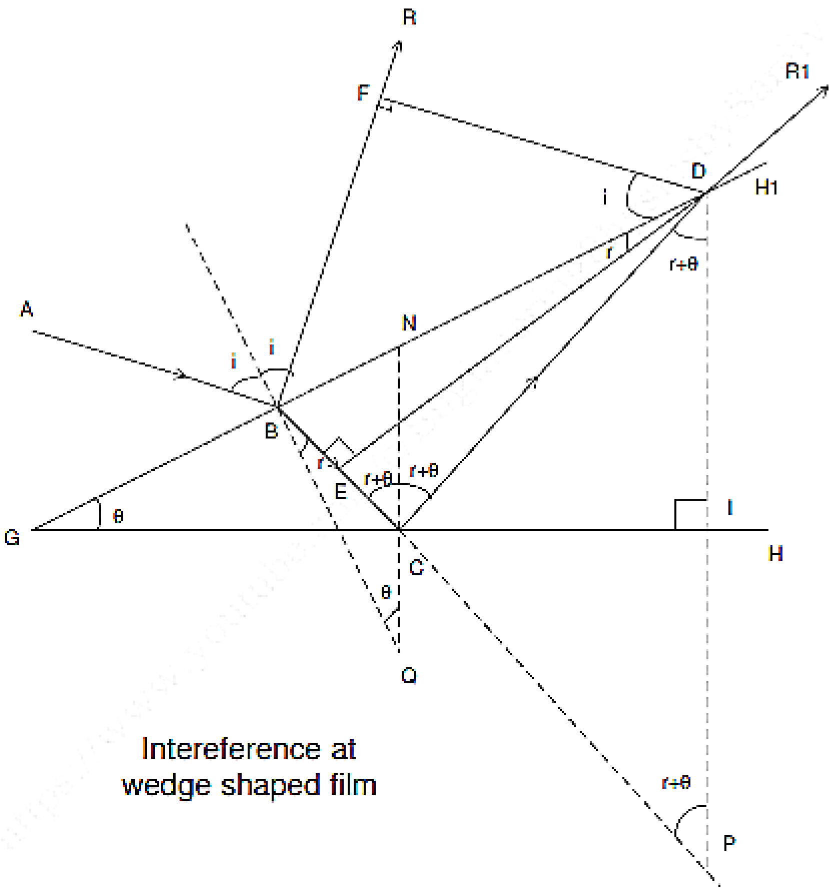
$$\Delta = \mu\,(BC + CD) - BF$$

$$\Delta = \mu\,(BE + EC + CD) - BF \text{-----------------------------------} (1)$$

$$\text{In } \Delta BDE,\ \sin r = \frac{BE}{BD} \quad \text{and} \quad \text{In } \Delta BFD,\ \sin i = \frac{BF}{BD}$$

$$\therefore \quad \mu = \frac{\sin i}{\sin r} = \frac{BF/BD}{BE/BD} = \frac{BF}{BE}$$

$$\therefore \quad BF = \mu\ BE \quad \text{-----------------------------------} (2)$$



Intereference at wedge shaped film

Substituting (2) in (1) and simplifying, we get

$$\boxed{\Delta = 2\,\mu\, t\, \cos(r+\theta)}$$

As the reflection at point B is at the surface of denser medium, additional path change of $\lambda/2$ is introduced for ray $BR_1$. Hence effective path difference becomes -

$$\boxed{\Delta = 2\,\mu\, t\, \cos(r+\theta) \pm \frac{\lambda}{2}}$$

# Path difference between two light waves in wedge film

$\Delta = 2\mu t \cos(r + \theta)$ .......... 1

Path difference due to reflection is

$\Delta = 2\mu t \cos(r + \theta) \pm \lambda/2$ ........... 2

## 1. Conditon for constructive interference/ maxima:

Path difference between two waves is integral multiple of λ

$$\boldsymbol{\Delta = n\lambda} \quad \textbf{............ 3}$$

Equating equation 2 & 3 we can write

$$2\mu t \cos(r + \theta) + (\lambda/2) = n\lambda$$

$$\boldsymbol{2\mu t \cos(r + \theta) = (2n + 1)\lambda/2} \quad \textbf{......... 4}$$

## 2. Condition for destructive interference/ Minima :

Path difference between two waves is odd integral multiple of λ

$$\boldsymbol{\Delta = (2n+1)\lambda/2} \quad ........... 5$$

Equating equation 5 & 2 we get

$$\boldsymbol{2\mu t \cos(r + \theta) = n\lambda} \quad \textbf{.......... 6}$$

# Expression for fringe width

The fringe width is the distance between the $(n+1)^{th}$ and $n^{th}$ fringes.

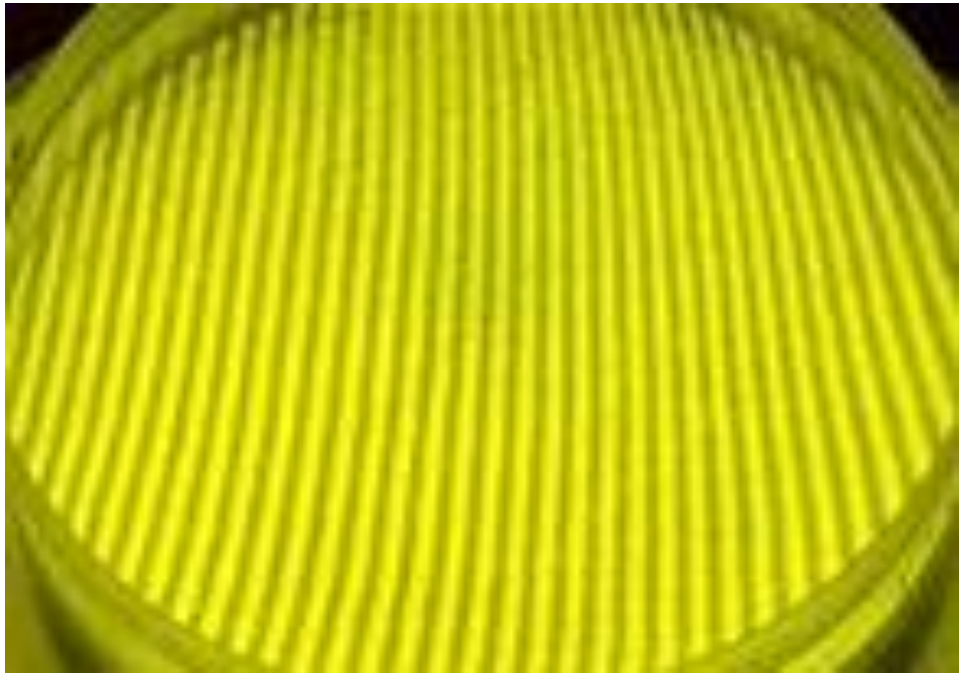
$$\beta = \frac{\lambda}{2\mu\Theta}$$

Where β is the fringe width

λ is the wavelength of light used

μ is the refractive index of the film

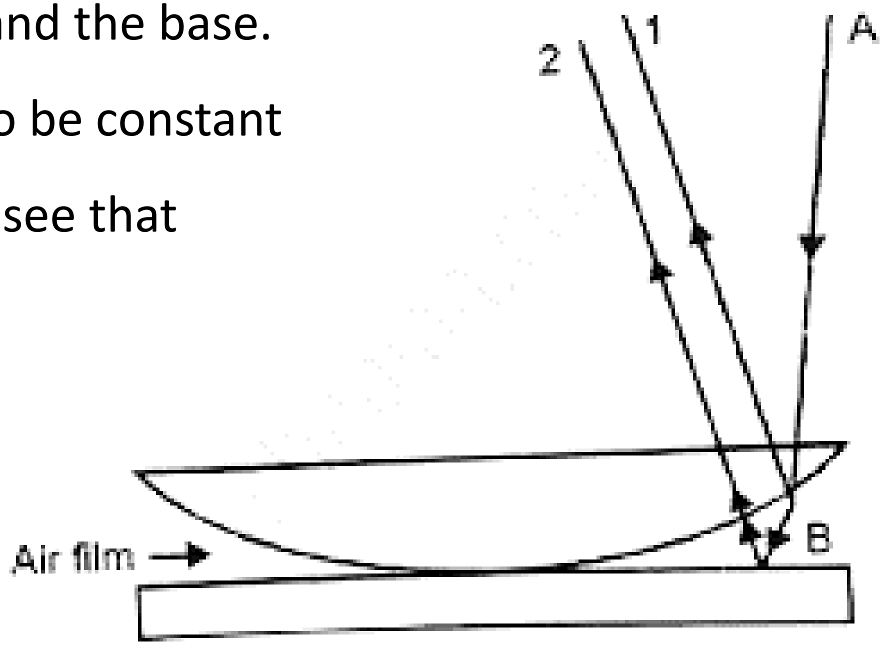
Θ is the wedge angle.



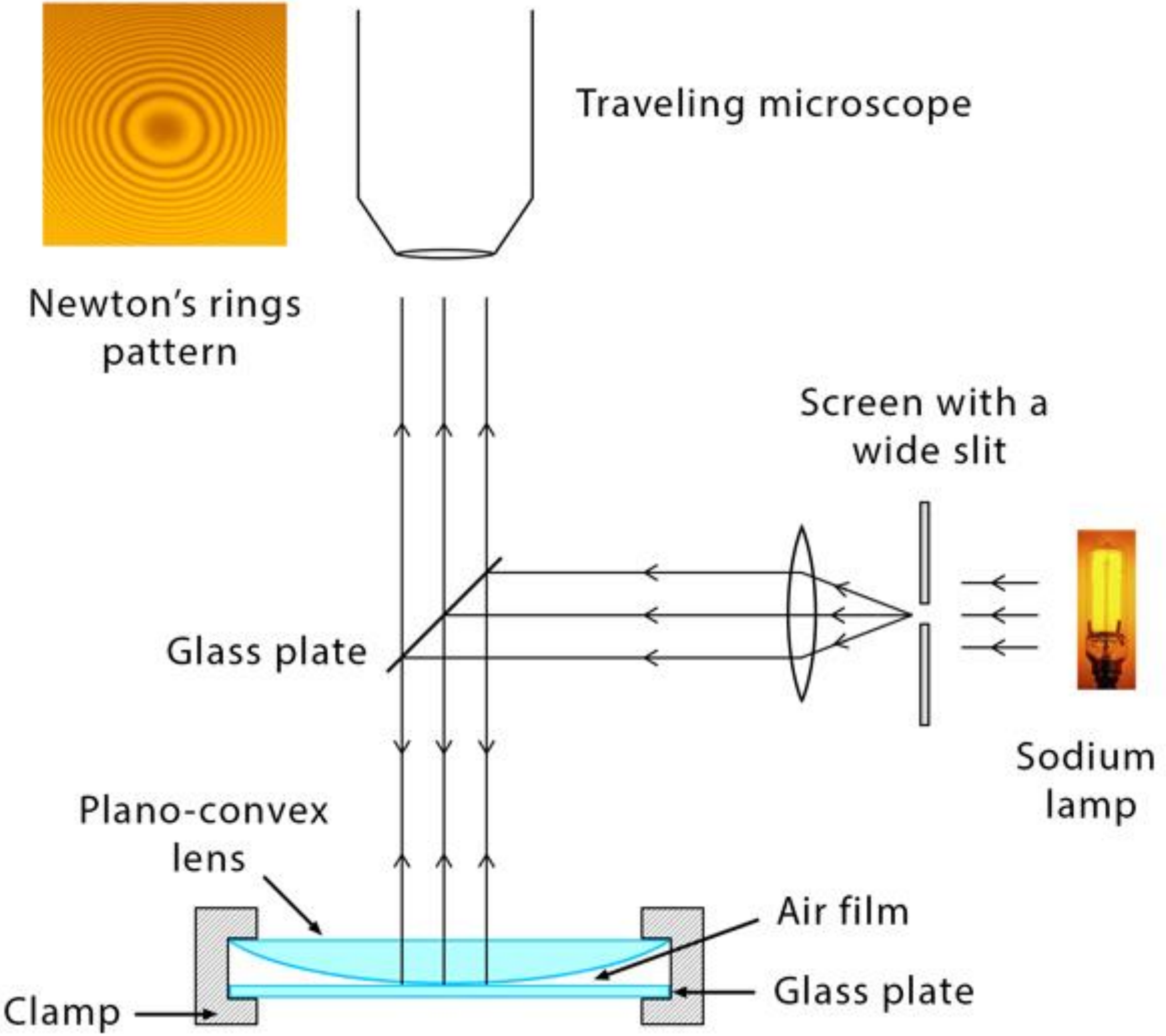
# Newton’s Rings

The rings of Newton's are formed as a result of interference which is between the light waves that are reflected from the top and bottom surfaces of the air film which is formed between the lens and glass sheet. A film of air which is of varying thickness is formed between the lens and the glass sheet.

The path of difference between the ray that is reflected and the ray which is the incident depends upon the thickness of the air gap between the lens and the base. As the lens is said to be symmetric along its axis the thickness is said to be constant along the circumference which is of a ring of a given radius. Hence we see that Newton's rings are circular.




Because of the thin air film formed between the glass plate and lens at the center, the central fringe of Newton's Rings is dark in the reflector system.


Traveling microscope
Newton's rings pattern
Screen with a wide slit
Glass plate
Sodium lamp
Plano-convex lens
Air film
Glass plate
Clamp

The path difference between the two successive reflected rays are obtained using wedge shaped interference case as:

$$\Delta = 2\mu t \cos(r+\theta) \pm \lambda/2 \qquad (1)$$

For air film, $\mu = 1$ and at normal incidence, $i = r = 0$. Since, $\theta$ is very small then $\cos\theta = 1$. Hence, eq. (1) reduces to

$$\Delta = 2t \pm \lambda/2 \qquad (2)$$

At the point of contact of the lens and the glass plate (O), the thickness of the film is effectively zero i.e. $t = 0$

$$\Delta = \pm\lambda/2 \qquad (3)$$

This is the condition for minimum intensity. Hence, the centre of Newton rings generally appears dark.

# DIAMETER OF NEWTON'S RING

Consider a ring of radius r due to thickness t of air film as shown in the figure

According to the geometrical theorem (i.e. property of the circle), the product of intercepts of the intersecting chord is equal to the product of sections of the diameter.

$$\overline{DB} \times \overline{BE} = \overline{AB} \times \overline{BC} \quad (4)$$

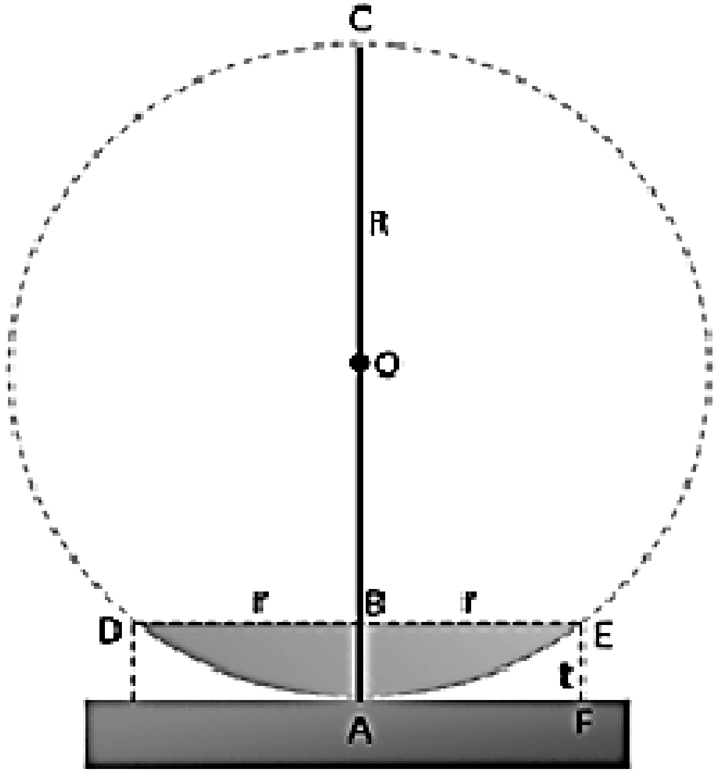
$$r \times r = t(2R - t) \quad (5)$$

$$r^2 = 2Rt - t^2 \quad (6)$$

Since $t$ is very small hence $t^2$ will also be negligible, thus,

$$r^2 = 2Rt \quad (7)$$

$$\boxed{t = \frac{r^2}{2R}} \quad (8)$$



a. ***Condition for a bright ring (constructive interference in thin film)***

$$2\mu t = (2n-1)\frac{\lambda}{2}$$ where $n$=1, 2, 3…..

$$\boxed{D_n = \sqrt{2(2n-1)\frac{\lambda R}{\mu}}}$$ where $n$ =1, 2, 3…..

b. ***Condition for a dark ring (destructive interference in thin film)***

$$2\mu t = n\lambda$$ where $n$ = 0,1, 2, 3…..

$$\boxed{D_n = \sqrt{\frac{4n\lambda R}{\mu}}}$$ where $n$ = 0, 1, 2, 3…..

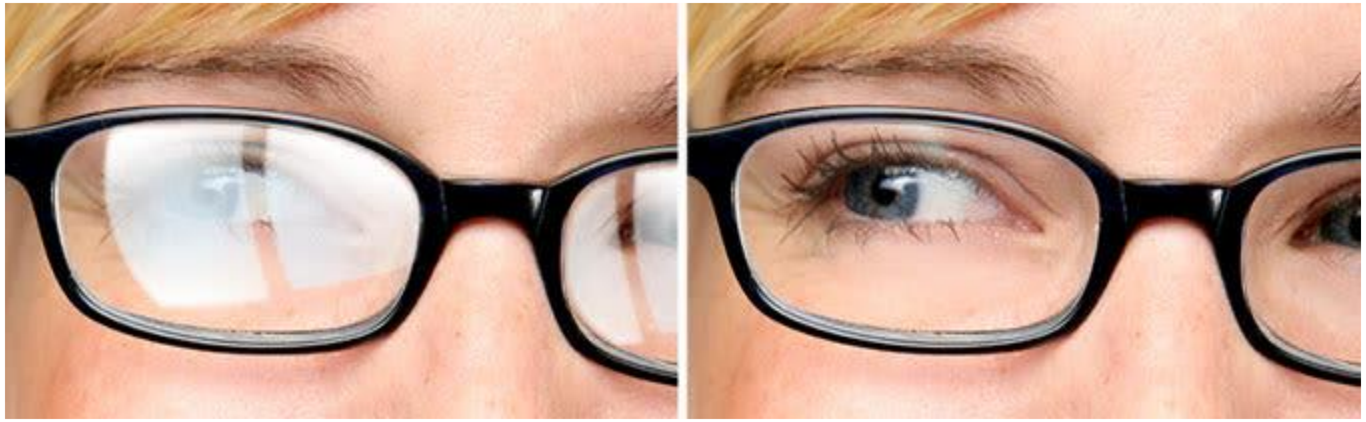
# Wavelength determination

The wavelength λ of light is given by the formula:

$$\lambda = \frac{D^2_{n+m} - D^2_n}{4mR}$$

Where, $D_{n+m}$ = diameter of (n+m)th ring,
$D_n$ = diameter of nth ring,
m = an integer number (of the rings)
R = radius of curvature of the curved face of the plano-convex lens.

# Anti reflection coating




# Practice questions

Suppose, in a given case, a thin film of oil (refractive index = 1.4) is spread on a surface of **water (refractive index = 1.3)**. Consider the oil to be spread uniformly over water surface in such a way that a **uniformly-thin film of oil** is formed as a layer over the water surface. What is the minimum possible thickness of the film if light of wavelength 560 nm is **strongly reflected** in case of **normal incidence**?

[**Hint 1**: Air-oil-water as consecutive media in given case.

**Hint 2:** Stokes' law may be considered in given case of thin film interference.

**Hint 3:** The medium above thin oil-film may be considered as air-medium of refractive index approximately 1

**Hint 4:** Strongly reflected light may be considered because of constructive interference between waves reflected from upper and lower surfaces of oil-film.]

. Assume that a student studies the formation of Newton's rings using a plano-convex lens. In this process, the student obtains the **diameter of 25th dark ring as 0.4 cm** and the **diameter of 40th dark ring as 0.7 cm.** If the radius of curvature of the plano-convex lens is considered to be 1 m, **calculate the wavelength** of monochromatic light used.

[If you have some time left after answering all of above mentioned questions; can you suggest/comment on the colour of the monochromatic light used in this study ?]

(**Hint**: Suppose the refractive index of thin-film between plano convex lens and glass-plate to be approximately 1)

# Thank you